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Nizami Gencevi Caddesi, No: 15

Söğütözü Mahallesi, Çankaya

Ankara, Türkiye

PERSONAL SUMMARY

Physics undergraduate at *METU* specializing in Optics and Photonics, skilled in optical system design, data analysis, and laboratory instrumentation. Awarded *TÜBİTAK* merit scholarship for academic performance, coordinating an alumni network and guiding prospective students toward higher education. Seeks a PhD in Physics to advance research in quantum optics.

EDUCATION

Middle East Technical University (METU) | Ankara, Türkiye Aug 2022 – Present

Bachelor of Science in Physics (Concentration: Applied Physics – Optics and Photonics)

- CGPA: 2.83/4.00 | Language of Instruction: 100% English
- Core coursework: Quantum Mechanics, Elementary Solid State, Laser Applications, Optics & Waves Lab, Electronic Lab, Modern Physics Lab, Computational Physics Lab

Vocational Open Education High School | Türkiye Aug 2021 – Jun 2022

- High School Diploma | GPA: 94.65/100.0 (High Honor Graduate) – accomplished via self-directed learning

Adıyaman Imam Hatip High School | Adıyaman, Türkiye Aug 2018 – Aug 2021

- Established school's first radio channel; hosted and produced content broadcast across social media platforms reaching schoolwide audience during pandemic

RESEARCH & TECHNICAL EXPERIENCE

Modern Physics Laboratory | METU Physics Department, Ankara, Türkiye Feb 2026 – June 2026

- Investigated wave-particle duality and quantized energy states through electron diffraction and Franck-Hertz experiments
- Explored the interaction of photon and matter by photoelectric effect, extracting work functions, and sodium atomic spectra
- Analyzed radioactive decay by alpha spectra and Compton scattering to investigate nuclear radiation properties
- Measured Hall coefficient, carrier sign, and drift mobility in Ge to determine carrier type and concentration

Independent Research – Quantum Optics & Photonics | Self-directed Dec 2025 – Present

- Studying on theoretical proposals of long-range quantum illumination radar for defense and sensing applications
- Investigating neutrino-based passive sensing systems for strategic assets detection and reactor monitoring applications

Optics Laboratory | METU Physics Department, Ankara, Türkiye Sep 2024 – Jan 2025

- Resolved atomic emission spectra using prism and grating spectrometers and extracted Cauchy coefficients of optical glass

- Explored wave optics through He-Ne laser wavelength measurement via Michelson interferometer and far-field single/slit diffraction interference envelopes
- Studied polarization and reflection properties including Fresnel reflection equations across s- and p-polarization states, Brewster angle, optical rotatory power of chiral molecules and confirmed Malus' law quantitatively
- Investigated optical properties through measuring bandpass and longpass filter transmittance spectra via UV-Vis spectrophotometry, converging and diverging lens systems, determining microwave propagation velocity in a dielectric medium

LEADERSHIP & VOLUNTEER EXPERIENCE

Peer Advisor & Student Orientation Volunteer | *METU, Ankara, Türkiye* Sep 2023 – Present

- Guide over 50 university students annually for their new university life
- Organize orientations and provide psychological and logistical support for first-year students
- Organized daily catering through the month of Ramadan for more than 1,000 students at Campus Mosque

Alumni Community Co-Leader | *Adıyaman, Türkiye, Hybrid* July 2023 – Present

- Mentor high school students preparing for National University Exams
- Manage a network of over 150 alumni for mentorship and career connections
- Organize seminars and events bridging high school graduates with university students

Search and Rescue Volunteer | *IHH Humanitarian Relief Foundation* Aug 2023 – Dec 2023

- Completed professional urban search and rescue training after firsthand personal experience in 2023 Kahramanmaraş earthquake
- Trained in disaster management logistics and advanced urban rescue techniques as part of a team

HONORS, AWARDS, CERTIFICATES & SCHOLARSHIPS

TÜBİTAK 2205 Undergraduate Merit Scholarship Sep 2022 – Present

- Granted for recognition of academic excellence by *TÜBİTAK (Scientific and Technological Research Council of Türkiye)*

TÜBİTAK 2204-A Regional Award Aug 2021

- Conducted research on the effectiveness of distance education methods, developed a pilot project and established a YouTube channel for providing lectures to underserved students in the pandemic

SKILLS & COMPETENCIES

Programming & Software: Python (Scientific Computing – NumPy, SciPy, Matplotlib), MATLAB, ANSYS Zemax (Optical Design), Microsoft Office Suite

Laboratory Skills: Optics & photonics experiments, electronics, modern physics instrumentation, data analysis and statistical error treatment

Languages: Turkish (Native), English (Advanced), Arabic (Intermediate), Kurdish (Intermediate)

Sports & Activities: Licensed Judo Athlete, Swimming, Camping and Survival Skills

REFERENCES

Available upon request